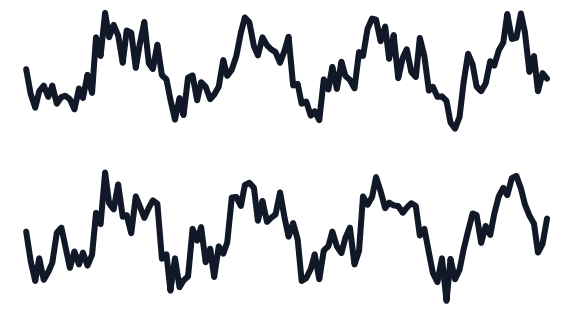


CoG-SAGE: Color-Order Guided, Session-Aware Gated EEG Decoding

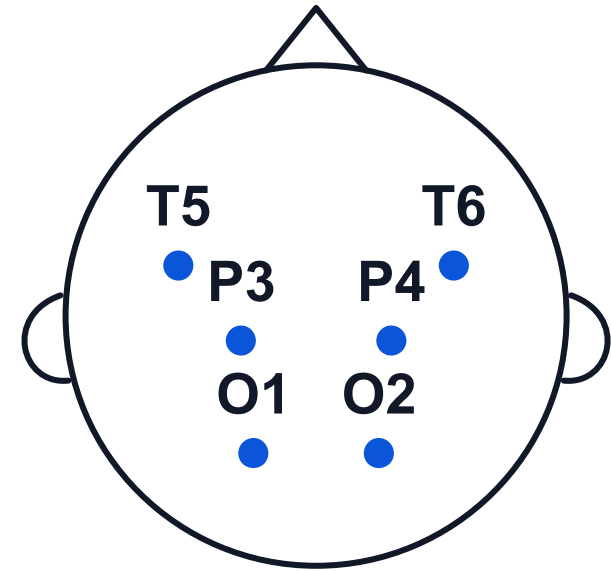
(a) EEG Input

Raw EEG epochs



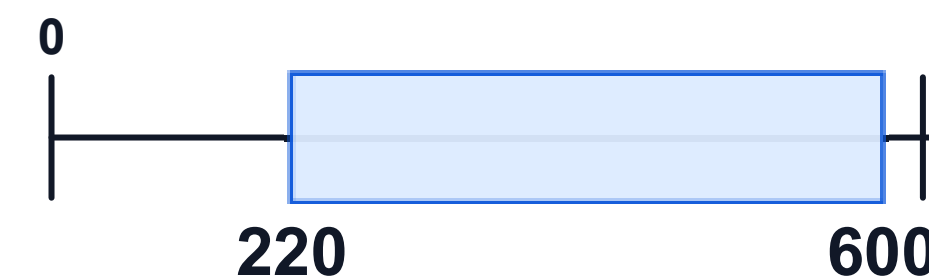
$$X \in \mathbb{R}^{B \times 6 \times 380}$$

Posterior 6 channels



Samples 220:600

(≈ 20 –400 ms post-stimulus)



7-class labels



Session-stratified split



(b) Preprocessing & Auxiliary Features

1) Raw normalization (train-only)

Train-only normalization of raw input (z-score, fit on training split only)

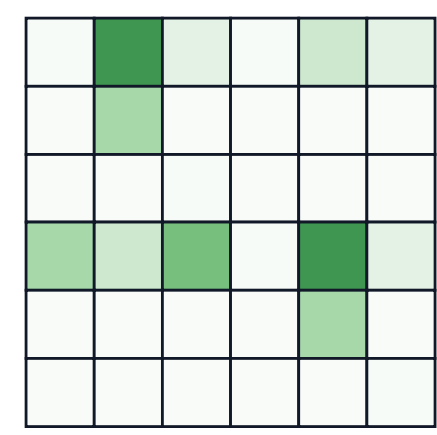


$$X_n \in \mathbb{R}^{B \times 6 \times T}$$

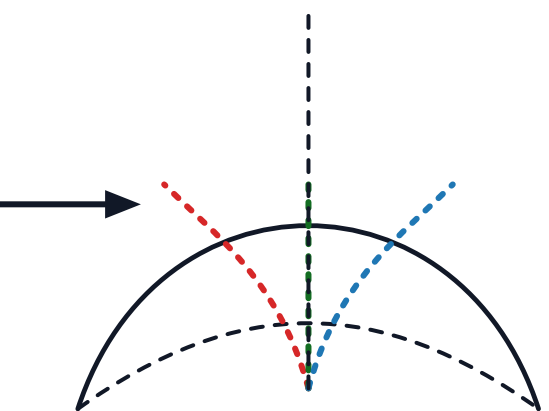


2) Auxiliary feature branch

(i) Covariance / Riemann auxiliary features



Covariance (6×6)



Log-Euclidean / Riemann mapping

(ii) Train-only normalization of auxiliary features

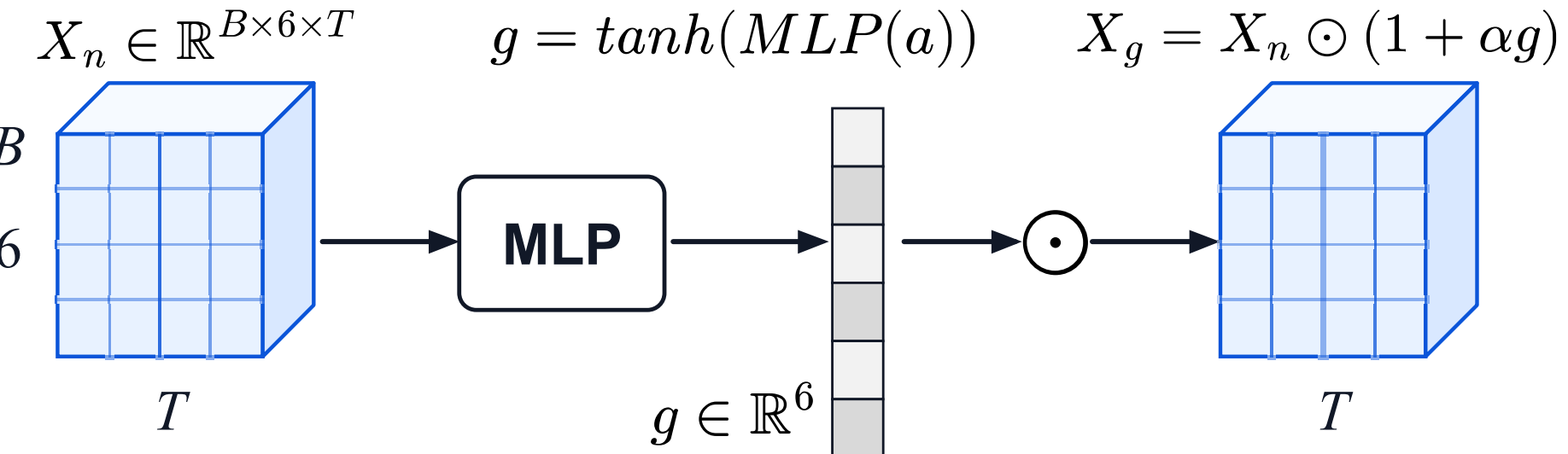
aux z-score, fit on training split only



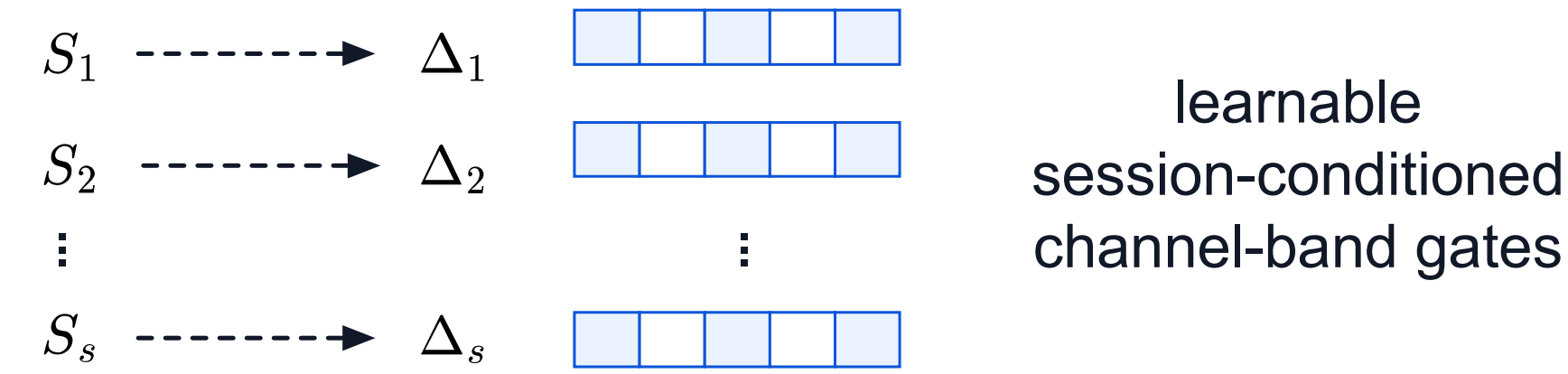
$$a \in \mathbb{R}^{d_{aux}}$$

(c) Representation Learning

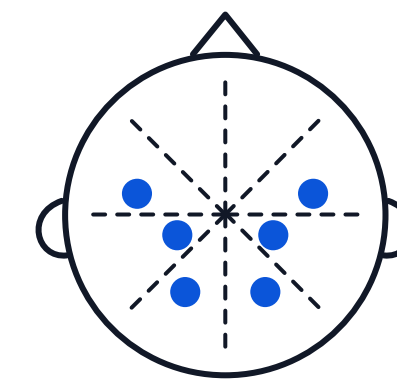
C1 Auxiliary-guided channel/spatial gating



C2 Session-aware SRM (channel-band spectral modulation)

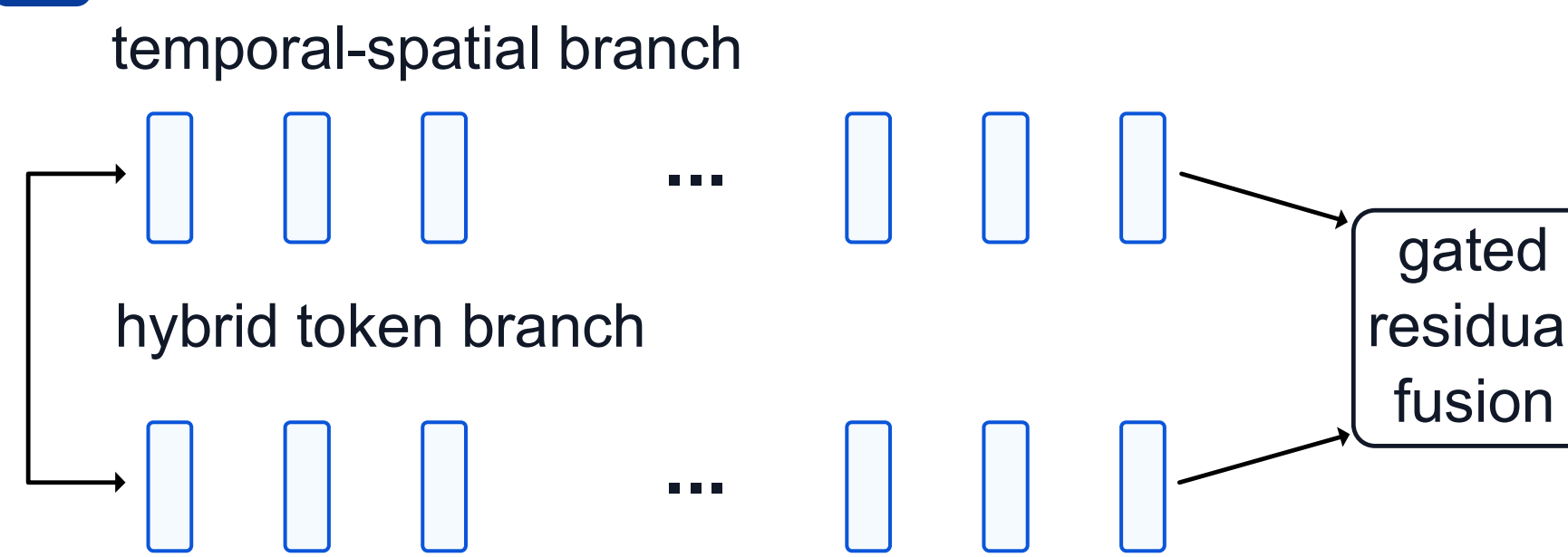


C3 Session-adaptive CAR front-end



learnable reference scale

C4 Hybrid raw backbone



C5 Fused representation

$$z \in \mathbb{R}^{d_z}$$



(d) Prediction Heads

Main classifier – 7-class logits



HSV prototype head – prototype similarities / logits 7-class outputs



Gated adjacent class-query head – residual 7-class logits



Feature-state residual head – residual 7-class logits



Combined logits

$$l = l_{main} + \lambda_{HSV} l_{HSV} + \lambda_{CQ} l_{CQ} + \lambda_{FT} l_{FT}$$

(e) Training Objective

Loss components

Adjacent Soft Focal Loss on combined logits

Center Loss on z

Color Contrastive Loss on representation / color embedding

Total objective

$$L = L_{ASF}(l, y) + \lambda_c L_{center}(z, y) + \lambda_{col} L_{contrastive}(\varphi(z), y)$$

Training strategies:

Mixup, EMA, Checkpoint averaging

Raw EEG flow

Auxiliary feature flow

Session-conditioning flow

Loss / gradient flow